Darshan UNIVERSITY

योग: कर्मसु कौशलम्

# Python Programming - 2301CS404

# Lab - 4

Roll No.:495

Name:Dhruv Kakkad

24031701022

## 01) WAP to print 1 to 10.

In [3]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

## 02) WAP to print 1 to n.

In [4]:
```
n = int(input("Enter a Number:"))
i=1
while i<=n:
    print(i)
    i+=1
```

```
Enter a Number: 10
```

```
1
2
3
4
5
6
7
8
9
10
```

## 03) WAP to print odd numbers between 1 to n.

In [7]:
```
n = int(input("Enter a Number:"))

i=1
while i<=n:
    if i%2!=0:
        print(f"Odd Number:{i}")
    i+=1
```

```
Enter a Number: 5
Odd Number:1
Odd Number:3
Odd Number:5
```

## 04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

In [ ]:
```
n = int(input("Enter 1st Num:"))
m = int(input("Enter 2nd Num:"))

while n<=m:
    if n%2==0 and n%3!=0:
        print(n)
    n+=1
```

```
Enter 1st Num: 5
Enter 2nd Num: 10
8
10
```

## 05) WAP to print sum of 1 to n numbers.

In [69]:
```
n = int(input("Enter a Number:"))
i=1
sum = 0
while i<=n:
    sum +=i
    i+=1
print(f"Sum of {n} number:{sum}")
```

```
Enter a Number: 10
Sum of 10 number:55
```

## 06) WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

In [8]:
```
n = int(input("Enter a Number:"))

i=1
sum=0
while i<=n:
    sum+=(i*i)
    i+=1
print(sum)
```

```
Enter a Number: 10
385
```

## 07) WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

In [14]:
```
n = int(input("Enter a Number:"))
i=1
sum=0
while i<=n:
    if i%2==0:
        sum-=i
    else:
        sum+=i
    i+=1
print(f"Sum of Series:{sum}")
```

```
Enter a Number: 10
Sum of Series:-5
```

## 08) WAP to print Multiplication Table of the given number.

In [16]:
```
n = int(input("Enter a Number:"))

i=1
while i<=n:
    print(f"{n} x {i} = {n*i}")
    i+=1
```

```
Enter a Number: 10
10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
```

## 09) WAP to find Factorial of the given number.

In [72]:
```
n = int(input("Enter a Number:"))
i=1
f=1
while i<=n:
    f=f*i
```

```
    i+=1
print(f"Factorial of a Number:{f}")
```

```
Enter a Number: 5
Factorial of a Number:120
```

## 10) WAP to print GCD of given two numbers.

In [73]:
```
n = int(input("Enter n:"))
m = int(input("Enter m:"))
min = n if n<m else m
i=1
while i<=min:
    if n%i==0 and m%i==0:
        gcd = i
    i+=1
print(f"GCD of a Number:{gcd}")
```

```
Enter n: 10
Enter m: 15
GCD of a Number:5
```

## 11) WAP to find Factors of the given number.

In [33]:
```
n = int(input("Enter a Number:"))
i=1
while i<=n:
    if n%i==0:
        print(f"Factors:{i}")
    i+=1
```

```
Enter a Number: 10
Factors:1
Factors:2
Factors:5
Factors:10
```

## 12) WAP to find whether the given number is Prime or not.

In [67]:
```
n = int(input("Enter a Number:"))
flag=0
i=2
while i<n:
    if n%i==0:
        break
    i+=1

if flag==0:
    print(f"{n} is a Prime Number")
else:
    print(f"{n} is not a Prime Number")
```

```
Enter a Number: 5
5 is a Prime Number
```

## 13) WAP to print sum of digits of given number.

In [50]:
```
import math
n = int(input("Enter a Number:"))
sum=0
while n>0:
    r=n%10
    sum = sum + r
    n=math.floor(n/10)
print(f"Sum Of Digits :{sum}")
```

```
Enter a Number: 55
Sum Of Digits :10
```

## 14) WAP to check whether the given number is Palindrome or not.

In [54]:
```
import math
n = int(input("Enter a Number:"))
sum=0
temp=n
while n>0:
    r=n%10
    sum=sum*10+r
    n=math.floor(n/10)

if temp==sum:
    print(f"{temp} is a Palindrome Number")
else:
    print(f"{temp} is not a Palindrome Number")
```

```
Enter a Number: 121
121 is a Palindrome Number
```

## 15) WAP to check whether the given number is an Armstrong Number or not.

In [58]:
```
import math
n = int(input("Enter a Number:"))
temp=n
sum=0
while n>0:
    r=n%10
    sum=sum+(r*r*r)
    n=math.floor(n/10)

if temp==sum:
    print(f"{temp} is a Armstrong Number")
else:
    print(f"{temp} is not a Armstrong Number")
```

```
Enter a Number: 153
153 is a Armstrong Number
```

## 16) WAP to print all the perfect numbers between 1 to n.

In [65]:
```
n = int(input("Enter a Number:"))
temp=n
i=1
```

```python
li = []
for i in range(1,n+1):
    sum = 0
    for j in range(1,i):
        if i%j==0:
            sum+=j
    if(sum==i):
        li.append(i)
print(li)
```

```
Enter a Number: 100
[6, 28]
```